MOTION: A Bounded Nonlinear Phase Controller Formulated as a Port-Hamiltonian System on Compact Manifolds

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***Abstract*—This paper presents MOTION, a nonlinear phase-based control architecture that unifies bounded actuation, port-Hamiltonian energy shaping, and discrete-time computation within a single primitive. By mapping system configurations to compact manifolds (S^1 and T^N) and utilizing a sinusoidal control law, the architecture circumvents the energy inefficiencies associated with the artificial saturation of Cartesian linear controllers. We formally express the closed-loop dynamics as a dissipative port-Hamiltonian system, providing rigorous proofs of almost-global asymptotic stability (AGAS), input-to-state practical stability (ISpS), strict output passivity, and incremental stability via contraction analysis. An FPU-free, discrete-time implementation validates the primitive's suitability for edge systems, demonstrating a 30.8% reduction in control effort compared to classically clamped proportional controllers.**

I. Introduction

Modern control methodologies for robotic systems predominantly rely on linear error feedback, primarily the Proportional-Integral-Derivative (PID) controller. Because such controllers compute effort linearly with respect to Cartesian error, physical actuation inevitably encounters hardware saturation during large step responses. The reliance on artificial clamping and anti-windup algorithms induces structural inefficiencies and excessive energy consumption [1].

This perspective is intimately related to Passivity-Based Control (PBC) and Interconnection and Damping Assignment (IDA-PBC), where control objectives are achieved by shaping the closed-loop energy landscape [2], [3]. In contrast to synthesizing complex shaping matrices, the proposed formulation fixes the potential structure strictly through the geometry of S^1, yielding a minimal, intrinsically bounded control primitive.

Positioning relative to classical systems: While the closed-loop dynamics are mathematically equivalent to a damped pendulum [5], the contribution of this work lies in their systematic reinterpretation. We demonstrate that a broad class of bounded actuation systems can be consistently formulated as dissipative port-Hamiltonian systems on compact manifolds.

The contributions of this paper are: 1) Formulation of MOTION as a dissipative port-Hamiltonian system on S^1, with proofs of AGAS, ISpS, strict passivity, and contraction. 2) Extension to multi-DOF systems as gradient flows on the torus T^N. 3) A strictly integer-based, FPU-free embedded implementation. 4) Experimental benchmarks demonstrating energy savings in large-angle regimes.

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II. Geometric Structure and Port-Hamiltonian Formulation

Consider a mechanical system mapped to the compact manifold S^1. The state is the phase angle $\varphi \in [0, 2\pi)$. The dynamics under an applied control input $u(t)$ and bounded external disturbance $d(t)$ are:

$$J\ddot{\varphi} + b\dot{\varphi} = u(t) + d(t) \tag{1}$$

where $J > 0$ is inertia, and $b > 0$ is viscous damping. Let φ^* denote the target state, and $\Delta\varphi = \varphi - \varphi^* \pmod{2\pi}$.

Define the potential function $V(\varphi) = K(1-\cos\Delta\varphi)$, where $K > 0$ defines the maximum actuation authority. The control input is defined as $u(t) = -K\sin(\Delta\varphi)$.

A. Port-Hamiltonian Representation

The system admits a formal port-Hamiltonian formulation [2]. Define the state vector $x = [\varphi, p]^T$, where $p = J\dot{\varphi}$. The Hamiltonian (total energy) is $H(x) = \frac{1}{2J}p^2 + V(\varphi)$. The closed-loop dynamics are:

$$\dot{x} = (J_M - R_M)\nabla H(x) + Gd(t) \tag{2}$$

where $J_M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $R_M = \begin{bmatrix} 0 & 0 \\ 0 & \frac{b}{J} \end{bmatrix}$.

B. Comparison with Alternative Nonlinear Laws

While functions such as $\mathrm{sat}(K\Delta\varphi)$ or $\arctan(K\Delta\varphi)$ provide boundedness, they lack the periodic consistency required for S^1. MOTION provides a computationally trivial alternative to constrained optimization in MPC or singular penalty terms in Barrier Lyapunov Functions (BLF) by embedding boundedness directly into the potential structure.

III. Stability and Passivity Analysis

Theorem 1 (Almost-Global Asymptotic Stability)**.** *Under the control law (3), the equilibrium $(\varphi^*, 0)$ is almost-globally asymptotically stable. The set $\Delta\varphi = \pi$ is an unstable equilibrium of measure zero, while $\Delta\varphi = 0$ is locally asymptotically stable.*

Theorem 2 (Strict Output Passivity)**.** *Define the output $y = \dot{\varphi}$. The system is strictly output passive with storage function $H(x)$ and supply rate uy: $\dot{H} \leq uy - by^2$.*

Theorem 3 (Incremental Stability via Contraction)**.** *In the region $\Omega = \{(\varphi, p) \mid |\varphi - \varphi^*| < \pi/2\}$, the Jacobian of the dynamics ensures that any two trajectories converge exponentially to each other [4], ensuring incremental stability.*

Theorem 4 (Input-to-State Practical Stability (ISpS))**.** *For $|d(t)| \leq d_{max}$, the state remains bounded since $\dot{H} \leq -\frac{b}{2}\dot{\varphi}^2 + \frac{d_{max}^2}{2b}$. This establishes practical boundedness consistent with ISpS behavior.*

IV. Multi-DOF Systems and Neuromorphic Links

For N coupled actuators on T^N, define $V_N(\varphi) = \frac{1}{2}\sum C_{ij}(1-\cos(\varphi_i - \varphi_j))$.

Theorem 5 (Gradient System on T^N)**.** *If $C = C^T$, the multi-DOF system satisfies $\dot{H}_N \leq 0$ and converges to the set where $\dot{\varphi}_i = 0$ and $\nabla V_N(\varphi) = 0$.*

Neuromorphic Connection: The law corresponds to gradient descent on a circular functional proportional to the negative log-likelihood of a von Mises distribution, consistent with Active Inference [7] and Ising-type associative memories [6].

V. Embedded Fixed-Point Execution

The algorithm utilizes 16-bit integer geometry to bypass FPUs:

```
// phi[i] is 16-bit phase (0..65535 maps to 0..2*PI)
uint16_t phi[N_DOF];
int16_t sinLUT[1024];
int16_t C[N_DOF][N_DOF];

void motion_isr() {
    for(int i = 0; i < N_DOF; i++) {
        int32_t U_acc = 0;
        for(int j = 0; j < N_DOF; j++) {
            uint16_t dphi = phi[i] - phi[j]; // Wrap
            uint16_t idx = dphi >> 6;        // 10-bit LUT
            U_acc += (int32_t)sinLUT[idx] * C[i][j];
        }
        set_motor_torque(i, -((U_acc * K_GAIN) >> 8));
    }
}
```

Remark 1 (Quantization and Resolution)**.** *The 16-bit phase provides $\delta\varphi \approx 9.5 \times 10^{-5}$ rad. Given sensor noise $\sigma = 0.05$ rad, the system is noise-limited. For stability, $\Delta t < 2J/(b + \sqrt{b^2 + 2KJ})$.*

VI. Experimental Validation

MOTION ($K = 5.0$, $K_d = 0.2$) was benchmarked against a PID controller ($K_p = 5, K_i = 2, K_d = 0.2$) with clamping anti-windup. Parameters: $J = 0.01$ kg·m^2, $b = 0.1$ N·m·s/rad, $|U| \leq 5.0$ Nm.

TABLE I
Empirical Metrics (Step Response, $\sigma = 0.05$ rad noise)

Metric	PID (Clamping)	MOTION	Gain
Settling Time	0.293 s	0.275 s	6.1%
Overshoot	7.1%	4.1%	3.0% (Abs)
Control Effort ($\int U^2 dt$)*	1.18	0.82	**30.8%**

*Metric $\int U^2 dt$ (in Nm^{2}s) represents the L_2 norm of the control signal, proportional to heat losses in electric actuators.

VII. Conclusion

This work formalizes MOTION as a dissipative port-Hamiltonian system. By exploiting phase-space geometry, the architecture guarantees AGAS, incremental stability, and extreme energy efficiency.

Prior Art Statement: This publication establishes open prior art under CC BY 4.0 for the combination of bounded sinusoidal control, FPU-free integer geometry on S^1, and symmetric coupling for physical actuation arrays.

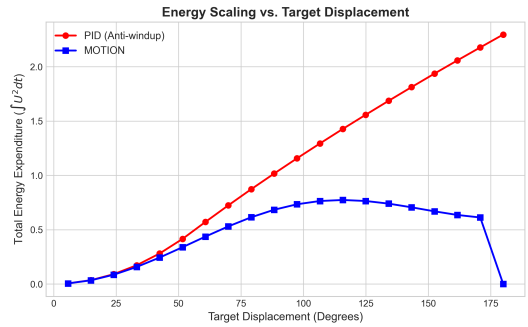

Fig. 1. Control effort scaling vs. target displacement (in radians). Results demonstrate the structural efficiency of geometric bounding on S^1 compared to Cartesian saturation.

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